Continued fractions as Möbius transformations

the trichotomy, and why $\begin{pmatrix}1 & 1\\ 0 & 1\end{pmatrix}$ is both the degenerate case and the alphabet

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1 A continued fraction is a composition of Möbius transformations

A *Möbius transformation* is a map

$$z \longmapsto \frac{az+b}{cz+d}, \qquad ad-bc \neq 0,$$

of $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ to itself. Substituting one into another and clearing denominators shows that *composition corresponds to matrix multiplication*: the assignment

$$\begin{pmatrix}a & b\\ c & d\end{pmatrix} \longmapsto \left(z \mapsto \frac{az+b}{cz+d}\right)$$

is a group homomorphism $GL_2(\mathbb{C}) \to$ Möb. Its kernel is the scalar matrices, since λ cancels from $(\lambda az + \lambda b)/(\lambda cz + \lambda d)$. So Möbius transformations are PGL_2, and any invariant of a transformation must be invariant under both conjugation and rescaling.

Now take the continued fraction of $\sqrt{7} = [\,2; \overline{1,1,1,4}\,]$, that is

$$\sqrt{7} \;=\; 2 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{4 + \ddots}}}}.$$

Every layer is the same map $f_a(z) = a + 1/z$, so the expansion is the composition $f_2 f_1 f_1 f_1 f_4 \cdots$. And $f_a(z) = (az+1)/(z+0)$, so

$$f_a \;\longleftrightarrow\; M(a) = \begin{pmatrix}a & 1\\ 1 & 0\end{pmatrix}.$$

That is where the $M(a)$ of `palindrome.tex` comes from: it is not a device, it is f_a written in coordinates. A continued fraction is a word in the alphabet $\{M(a)\}$, and the value $[a_0; a_1, \ldots]$ is the limit of $M(a_0)M(a_1)\cdots$ applied to ∞.

Periodicity is a fixed point

Fixed points come out of the definition. If $q \neq 0$, the fixed points of $A = \begin{pmatrix} p & p' \\ q & q' \end{pmatrix}$ are the roots of

$$q\,x^2 + (q' - p)\,x - p' \;=\; 0, \tag{1}$$

a quadratic whose discriminant is

$$(q' - p)^2 + 4qp' \;=\; (p + q')^2 - 4(pq' - p'q) \;=\; \operatorname{tr}(A)^2 - 4\det(A).$$

Write $\Delta(A)$ for this. Note that Δ is not quite the right object: rescaling A by λ multiplies it by λ^2. The genuine invariant of the *transformation* is $\operatorname{tr}(A)^2/\det(A)$, and normalising $\det A = 1$ leaves $\operatorname{tr}(A)^2$, which is why signs of traces never matter below.

For $\sqrt{7}$: the period is $1, 1, 1, 4$, and saying it repeats says that the complete quotient

$$\alpha_1 \;=\; \frac{1}{\sqrt{7} - 2} \;=\; \frac{2 + \sqrt{7}}{3}$$

is returned unchanged by $f_1 f_1 f_1 f_4$. That composite is

$$A \;=\; M(1)M(1)M(1)M(4) \;=\; \begin{pmatrix} 14 & 3 \\ 9 & 2 \end{pmatrix},$$

and (1) reads $9x^2 - 12x - 3 = 0$, that is $3x^2 - 4x - 1 = 0$, with roots $(2 \pm \sqrt{7})/3$. Its discriminant is $144 + 108 = 252 = 16^2 - 4$, in agreement with $\Delta(A) = \operatorname{tr}^2 - 4\det$.

> "$\sqrt{7}$ has period 4" and "$(2 + \sqrt{7})/3$ is a fixed point of $\begin{pmatrix} 14 & 3 \\ 9 & 2 \end{pmatrix}$" are the same sentence. Everything in this note follows from replacing the first formulation by the second and then asking the questions a Möbius transformation invites.

T is the alphabet

Proposition 1. *Let* $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ *and* $J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, *i.e.* $z \mapsto z + 1$ *and* $z \mapsto 1/z$. *Then*

$$M(a) \;=\; T^a J \qquad \textit{for every } a.$$

Proof. $T^a = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$, and right multiplication by J swaps columns. □

This is not a coincidence but a restatement of the algorithm. The continued fraction algorithm makes exactly two gestures: subtract the integer part, which is iterating T^{-1}, and invert the remainder, which is J. So T and J *are* the algorithm, in coordinates.

> $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is not a curiosity at the edge of the theory. It is the *alphabet*. Every continued fraction is a word in translations T^a interleaved with a single flip, and the period of $\sqrt{n}$ is one such word.

And yet T is also the one transformation the theory has nothing to say about, because $\Delta(T) = 0$. The rest of this note is that coincidence.

2 The trichotomy: the classification of Möbius transformations

The three cases below are not specific to continued fractions. They are the standard classification of Möbius transformations, obtained by conjugating to a normal form.

Proposition 2. *A Möbius transformation other than the identity has one or two fixed points. If two, conjugating them to* 0 *and* ∞ *puts it in the form* $z \mapsto kz$ *for a unique* $k \neq 0, 1$ *(up to* $k \leftrightarrow 1/k$*), and* $\mathrm{tr}^2 / \det = k + 2 + 1/k$*. If one, conjugating it to* ∞ *puts the map in the form*

$$z \longmapsto z + 1,$$

and $\mathrm{tr}^2 / \det = 4$.

So T is not an example of the degenerate case: T *is its normal form.* Every Möbius transformation with a single fixed point, anywhere, is conjugate to $z \mapsto z + 1$. Equivalently: "A is not diagonalisable" and "A has one fixed point" are the same statement, because diagonalisable means conjugate to $z \mapsto kz$, which means two fixed points.

Specialise to $A \in SL_2(\mathbb{Z})$, so $\det A = 1$ and $\Delta(A) = \mathrm{tr}(A)^2 - 4$. Because $\mathrm{tr}\, A$ is now an *integer*, exactly three cases arise.

$\lvert \mathrm{tr}\, A \rvert$	Δ	name	normal form	fixed points
> 2	> 0	hyperbolic	$z \mapsto kz$, $k > 0$	two, real quadratic, conjugate
$= 2$	$= 0$	parabolic	$z \mapsto z + 1$	one, rational (or ∞)
< 2	< 0	elliptic	$z \mapsto e^{i\theta} z$	two, complex conjugates

Each row is a statement about continued fractions, and the three statements together are the reason this project has the shape it has.

The same three cases as isometries of the hyperbolic plane

A Möbius transformation with real coefficients preserves the upper half-plane $\mathbb{H}$, and the maps so arising are exactly its orientation-preserving isometries. The trichotomy is then geometric, and this is where the names come from.

	as an isometry of $\mathbb{H}$	in the modular surface $\mathbb{H}/SL_2(\mathbb{Z})$
hyperbolic	translation along a geodesic axis	a closed geodesic
parabolic	shear fixing one boundary point, preserving horocycles; **no axis**	the cusp
elliptic	rotation about an interior point	the cone points, orders 2 and 3

The translation length ℓ of a hyperbolic isometry satisfies $\lvert \mathrm{tr} \rvert = 2\cosh(\ell/2)$. Proposition 4 below is that formula: the regulator $R(n)$ is *half the translation length* of the period word, and the period $L(n)$ counts how its axis crosses the tessellation of $\mathbb{H}$ by copies of the fundamental domain. For $\sqrt{7}$: $\mathrm{tr} = 16$, $R = \log(8 + 3\sqrt{7}) = 2.7687$, $\ell = 5.5373$, and $2\cosh(2.7687) = 16$.

That the coding of a geodesic by its crossings of the tessellation reproduces the continued fraction is Series' theorem. It is also why a ratio like L/R — crossings per unit hyperbolic length — converges to a constant at all: the geodesic flow on the modular surface is ergodic, and the Gauss map is its Poincaré section. §8 is about the correction term to that constant.

Hyperbolic: the periodic case

A has real eigenvalues $\varepsilon, 1/\varepsilon$ with $\varepsilon > 1$, and its fixed points are a conjugate pair of real quadratic irrationals. These are the numbers with an eventually periodic continued fraction. This row is Lagrange's theorem, and all of `period.py`, `palindrome.tex` and the 99,683 non-squares below 10^5 live here.

Parabolic: the case the project excludes

$\Delta = 0$ means (1) has a double root, which is rational. Up to sign, A is conjugate in $SL_2(\mathbb{Z})$ to a power of T, and *the fixed point is a cusp.*

Lemma 3. *For every $p/q \in \mathbb{Q}$ there is $G \in SL_2(\mathbb{Z})$ with $G \cdot \infty = p/q$, and then GTG^{-1} is parabolic and fixes p/q. Conversely every parabolic element of $SL_2(\mathbb{Z})$ other than $\pm I$ arises this way.*

Proof. Choose u, v with $pu + qv = 1$ and put $G = \begin{pmatrix} p & -v \\ q & u \end{pmatrix}$, of determinant 1. Conjugation preserves the trace, so $\operatorname{tr}(GTG^{-1}) = 2$ and $\Delta = 0$. For the converse, a parabolic element has a unique fixed point, rational by (1) with $\Delta = 0$; conjugate it to ∞ and a matrix in $SL_2(\mathbb{Z})$ fixing ∞ with trace 2 is T^k. □

Three worked cases from `matrices.py`:

$$\tfrac{0}{1}: \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}, \qquad \tfrac{3}{1}: \begin{pmatrix} -2 & 9 \\ -1 & 4 \end{pmatrix}, \qquad \tfrac{2}{5}: \begin{pmatrix} -9 & 4 \\ -25 & 11 \end{pmatrix},$$

all of trace 2.

This row is where the perfect squares went. Throughout this repository n is assumed to be a non-square; the 316 squares below 10^5 are silently discarded, with the reason usually given as "$\sqrt{n}$ is then rational". The matrix statement is sharper:

	hyperbolic	parabolic
fixed point	quadratic irrational	rational
continued fraction	infinite, periodic	finite; Euclid halts
eigenvalues	$\varepsilon, 1/\varepsilon$	$1, 1$
diagonalisable	yes	**no**
$\log \varepsilon$	$R(n) > 0$	0
unit rank of the field	1	0
Pell $x^2 - ny^2 = 1$	infinitely many solutions	only $(\pm 1, 0)$

T fails to be diagonalisable for exactly the reason the continued fraction terminates. In the hyperbolic case the two fixed points are α and $\bar{\alpha}$, and the algorithm is the passage between them; the palindrome argument is built on their being *distinct* and oppositely placed. Parabolic is the collision $\alpha = \bar{\alpha}$. A repeated eigenvalue with a one-dimensional eigenspace and a repeated root of the fixed-point equation are the same degeneracy, seen from two sides.

This also reads the other way. The obstruction in `cube-roots.tex` was that the two remaining conjugates of $\sqrt[3]{n}$ are complex, so there is no interval to confine $\bar{\alpha}$ to. Here the obstruction is that there is only *one* conjugate. Both are failures of the same requirement: two distinct real fixed points, comparably placed.

Elliptic: torsion, and the complete list of alternative digits

$|\operatorname{tr} A| \leq 1$ forces A to have finite order, and since $\operatorname{tr} A \in \{-1, 0, 1\}$ together with $\pm I$, the orders occurring in $SL_2(\mathbb{Z})$ are exactly $1, 2, 3, 4, 6$. Computed:

A	tr	Δ	order	fixed points in
I	2	0	1	—
$-I$	-2	0	2	—
$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$	0	-4	4	$\mathbb{Q}(i)$
$ST = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$	1	-3	6	$\mathbb{Q}(\omega)$
$(ST)^2 = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}$	-1	-3	3	$\mathbb{Q}(\omega)$

In $\mathbb{H}$ these are the rotations: S fixes i and ST fixes a primitive sixth root of unity, and modulo $\pm I$ they have orders 2 and 3 — the two cone points of the modular orbifold. The orders 4 and 6 in $SL_2(\mathbb{Z})$ become 2 and 3 in $PSL_2(\mathbb{Z})$, which is the group that actually acts, since scalars act trivially.

A complex number has no continued fraction over $\mathbb{Z}$ at all, so this row is invisible from inside the real theory. But it is the row that explains a choice made in `cube-roots.tex`.

The elliptic fixed-point fields of $SL_2(\mathbb{Z})$ are $\mathbb{Q}(i)$ and $\mathbb{Q}(\omega)$, and *nothing else*: the only negative values of $\operatorname{tr}^2 - 4$ over integer traces are -4 and -3. So Hurwitz's continued fraction over $\mathbb{Z}[i]$ (1887) and the Eisenstein continued fraction of `cube-roots.tex` §7 are not two ideas among many. They are the complete list of rings of digits that $SL_2(\mathbb{Z})$ offers. $\mathbb{Z}[i]$ and $\mathbb{Z}[\omega]$ are the only imaginary quadratic rings whose unit group is larger than $\{\pm 1\}$, which is the same statement as S having order 4 and ST order 6.

3 The trace is the regulator

Let L be the period of $\sqrt{n}$ and $A = M(a_1)M(a_2)\cdots M(a_L)$ the matrix of one period. Since $\det M(a) = -1$,

$$\det A = (-1)^L.$$

So A is in $SL_2(\mathbb{Z})$ exactly when L is even. The two cases are the two Pell equations, and they are also the two hyperbolic functions.

Proposition 4. *With (x, y) the fundamental solution of $x^2 - ny^2 = (-1)^L$ and $R = \log(x + y\sqrt{n})$,*

$$\operatorname{tr} A \;=\; 2x \;=\; \begin{cases} 2\cosh R, & L \text{ even } (\det A = +1), \\ 2\sinh R, & L \text{ odd } (\det A = -1). \end{cases}$$

Proof. The eigenvalues of A multiply to $\det A = (-1)^L$ and the larger is $\varepsilon = x+y\sqrt{n} = e^R$, because A fixes the complete quotient α_1 and acts on $\mathbb{Z}[\sqrt{n}]$ by multiplication by the fundamental unit. For L even the pair is (e^R, e^{-R}), summing to $2\cosh R$; for L odd it is $(e^R, -e^{-R})$, summing to $2\sinh R$. Either way the sum is $x + y\sqrt{n} + x - y\sqrt{n} = 2x$ on expanding $\varepsilon^{\pm 1}$. □

Verified exactly in `matrices.py` for n up to 1621 (period 79, trace a 39-digit integer):

n	L	$\det A$	$\operatorname{tr} A = 2x$	$x^2 - ny^2$
2	1	−1	2	−1
7	4	+1	16	+1
13	5	−1	36	−1
46	12	+1	48670	+1
61	11	−1	59436	−1
151	20	+1	3456296080	+1
421	37	−1	88084891393642836	−1

Three things this repository tracks separately are one thing. The *parity of the period*, the *solvability of $x^2 - ny^2 = -1$*, and the *determinant of the period word* are the same invariant, namely $(-1)^L$; and that invariant decides whether the trace is $2\cosh R$ or $2\sinh R$. The 11.52% of $n < 10^5$ with odd period reported in the README are exactly the n whose period word lies in $GL_2(\mathbb{Z}) \setminus SL_2(\mathbb{Z})$.

4 The palindrome as a statement about conjugacy

$M(a)$ is symmetric, so transposing a product reverses it:

$$\big(M(a_1)\cdots M(a_L)\big)^{\top} \;=\; M(a_L)\cdots M(a_1).$$

Hence $A^{\top}$ is the matrix of the reversed period, and the palindrome theorem says precisely that the reversed period is a cyclic rotation of the original — equivalently that A *is*

conjugate to $A^\top$ by a product of the $M(a_i)$. Checked for $n = 7, 13, 46, 151$ in `matrices.py`; e.g. $n = 7$, period $1, 1, 1, 4$:

$$A = \begin{pmatrix} 14 & 3 \\ 9 & 2 \end{pmatrix}, \qquad A^\top = \begin{pmatrix} 14 & 9 \\ 3 & 2 \end{pmatrix} = M(4)M(1)M(1)M(1).$$

Every matrix in SL_2 is conjugate to its transpose over the rationals, so the content is entirely in the conjugating element being *integral and a word in the same alphabet.* Lemma **??** of `palindrome.tex` — that the fixed points of A and $A^\top$ correspond under $x \mapsto -1/x$ — is the fixed-point shadow of this.

Hyperbolic conjugacy classes in $SL_2(\mathbb{Z})$ correspond to closed geodesics on the modular surface, of length $2R$; a class conjugate to its own transpose is what Sarnak calls a *reciprocal* geodesic, one invariant under an orientation-reversing involution. Legendre's palindrome theorem is the assertion that the principal cycle of discriminant $4n$ is always reciprocal.

5 Why one dimension is exactly the right number

The boxed claim at the end of `cube-roots.tex` was that a continued fraction "is a one-dimensional device … and it can express only a rank-one unit group". In the 2×2 picture that stops being a slogan.

Proposition 5. *Let $A \in SL_2(\mathbb{Z})$ be hyperbolic with fixed points generating the real quadratic field K and order $\mathcal{O}$. Then the centraliser of A in $SL_2(\mathbb{Z})$ is $\{\pm 1\} \times \langle A_0 \rangle \cong \mathbb{Z}/2 \times \mathbb{Z}$ for some A_0, and it is isomorphic to the norm-one unit group $\mathcal{O}^1$.*

The rank is 1 on both sides, and that matching is the whole reason the algorithm works. Dirichlet gives $\operatorname{rank} \mathcal{O}^\times = r_1 + r_2 - 1 = 1$ for a real quadratic field; the continued fraction produces a *cyclic* object, a period defined up to rotation, and a cyclic object has rank 1. Supply and demand agree.

field	unit rank	period must be	device
$\mathbb{Q}$ (n a square)	0	nothing; finite	parabolic, T
$\mathbb{Q}(\sqrt{n})$	1	a circle	continued fraction
$\mathbb{Q}(\sqrt[3]{n})$	1	a circle	open (Hermite)
$\mathbb{Q}(\omega, \sqrt[3]{n})$	2	a torus	none of dimension 1

Rows one and two are the trichotomy. Row three is the irritating one: the rank *is* 1, so a one-dimensional device is not ruled out by counting — which is why Hermite's problem is hard rather than impossible. What fails for row three is not the rank but the ordering, as `cube-roots.tex` §3 shows. Row four is ruled out by counting, and that is a genuine impossibility rather than an open problem.

6 A measured consequence: where the Lévy shortfall comes from

`analyse.py` predicts $L/R \to 1/\lambda$ with $\lambda = \pi^2/(12 \log 2) = 1.186569$, on the grounds that the complete quotients of a typical continued fraction are Gauss-distributed. The data

disagree by about 10%, and in the matrix language the reason is visible: the period word is not a typical word, because Proposition 1 plus the palindrome force its last letter to be $T^{2a_0}J$ with $2a_0 \approx 2\sqrt{n}$. One letter of the word is enormous by construction.

Its contribution to R is $\log(2a_0)/L$ per letter. Removing it:

band	R/L	excess over λ	$\log(2a_0)/L$	R/L less that letter
$[2, 10^3)$	1.5025	0.3160	0.6415	1.0084
$[10^3, 10^4)$	1.4861	0.2995	0.4418	1.1411
$[10^4, 10^5)$	1.4306	0.2440	0.2937	1.2094
$[10^5, 10^6)$	1.3275	0.1409	0.1536	1.2071

Two things to read off. First, the excess tracks $\log(2a_0)/L$ closely in the top band (0.141 against 0.154), so the single forced letter accounts for most of the discrepancy. Second, and this is the point for the README's claim that the law *drifts with* n: the excess decays like $\log n/L \sim \log n/\sqrt{n}$, which is slow. Part of the drift documented in `distribution.pdf` is therefore not a deep fact about the distribution of $L(n)/\sqrt{n}$ — it is one oversized partial quotient, mandated by the palindrome, contaminating the regulator at order $\log n/\sqrt{n}$.

Honesty about what remains: the trimmed value settles near 1.207, not $\lambda = 1.187$, a residual excess of 1.7%. The period's remaining quotients really are slightly larger than Gauss-typical. That is a separate effect and this note does not explain it.

7 Summary

	hyperbolic	parabolic	elliptic
$\Delta = \mathrm{tr}^2 - 4\det$	> 0	$= 0$	< 0
Möbius normal form	$z \mapsto kz$, $k > 0$	$z \mapsto z + 1$	$z \mapsto e^{i\theta}z$
isometry of $\mathbb{H}$	translation	shear at a cusp	rotation
model	$M(a)$ words	$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$	S, ST
fixed points	real quadratic	rational	$\mathbb{Q}(i)$, $\mathbb{Q}(\omega)$
continued fraction	infinite, periodic	finite	none over $\mathbb{Z}$
diagonalisable	yes	**no**	yes over $\mathbb{C}$
regulator	$R > 0$	0	—
unit rank	1	0	0
in this repository	the 99,683	the 316 squares	`cube-roots` §7

The answer to "what about $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$" is that it occupies both ends of the theory at once. It is the degenerate case, the one matrix with a repeated eigenvalue, where the period collapses and Euclid halts. And it is the generator: $M(a) = T^a J$, so every period of every $\sqrt{n}$ is spelled in it. The theory of periodic continued fractions is the study of which *words* in T are hyperbolic, conducted in a language whose only letter is not.